

Chyi Ricketts

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EDUCATION

Imperial College London

London, UK

Anticipated Sept. 2026

Masters of Research (MRes) in Cancer Informatics

- **Modules:** The Molecular Basis of Cancer and Cancer Informatics (R/Unix, Python, Linux, and NGS)
- **Project:** 10-Month Dedicated Research Project in ML Algorithms for Drug Discovery

Duke University, Duke Kunshan University Dual Degree

May 2025

Durham, NC, USA

Bachelor of Science in Molecular Biosciences with a Track in Genetics and Genomics

Bachelor of Science in Interdisciplinary Studies (Subplans: Molecular Biosciences; Genetics and Genomics)

Honors: Graduation with Distinction

- **Cumulative GPA:** 3.7/4.0
- **GRE:** Verbal 160 | Quantitative: 167
- **Relevant Coursework:** Computational Genomics, Computational Biology & Bioinformatics, Advanced Methods in Functional Genomics

Hunterdon County Academics, Biomedical Sciences Academy

May 2021

Annandale, NJ, USA

Dual Enrollment with Rutgers University for courses in Medical Dynamics, Anatomy and Physiology I & II, and Medical Terminology

- Cumulative GPA: 3.9/4.0
- **Awards and Honors:** John Philip Sousa Band Award, National Honor Society, Science HS, Math HS
- **Extracurriculars:** Drum Major/Section Leader of Marching Band for 2 years, NJ Youth Symphony

RESEARCH EXPERIENCE

Ballester Lab Master's Student in Cancer Informatics

Dec. 2025 – Sep. 2026

- Building improved target-specific machine-learning scoring functions for each of five cancer targets
- Methods using Graph neural networks: D-MPNN, GLAM, GEM, Featurization schemes AEV, PLEC, GRID, and benchmarking protocols: splits by various molecular dissimilarity metrics

Huang Lab Student Research Scholar

May 2023 – Present

Kunshan, Jiangsu, China

Thesis: Integrating Segmentation Software and BLIP-2 for Automated Cellular Data Analysis in a Web-Based Application

- Creation of an original, user-friendly web application for image visualization: <https://sw-cellvis.onrender.com/>
- Literature Review evaluating 8 scientific image analysis software programs, quantitatively benchmarking 3 ML or DL -based Cellular Segmentation Software Programs (Ilastik, Cellpose, and Segment Anything)

Project: RNAi screen to identify novel genes involved in cancer cell proliferation

- Streamlined the extraction and purification of prokaryotic siRNA followed by transfection of siRNA into HeLa cells to evaluate the extent of protein expression and toxicity

Grant writing: Authored and submitted a successful grant proposal to the Kunshan Supercomputing Center, resulting in the award of 9 high-performance computing nodes to support image analysis and segmentation software benchmarking

Di Talia Lab & Bagnat Lab Undergraduate Research Assistant in Computational Analysis

March 2024 – July 2024

Durham, NC, USA

Project: DST (Duke Science and Technology) SPARK Project: “Synchronized Clocks in Zebrafish Patterning”

- Trained 10 Cellpose models and quantitatively evaluated their performance using custom Python metrics
- Confocal Image Analysis with DL methods to study the oscillating ERK expression in embryonic zebrafish notochords
- Handling developing zebrafish and imaging notochords 1 to 8 days post-fertilization with a confocal microscope

IRB Barcelona – Institute for Research in Biomedicine Research Assistant

Oct. 2022 – Dec. 2022

Barcelona, Spain

- Evaluated the extent of cell toxicity for varying doses of up to 8 cancer drugs on cells
- Assisted in procedures that involved handling live mice to study stem cells and its relation to cancer

Bagnat Lab Undergraduate Research Assistant in Molecular Biology

April 2022 – June 2022

Durham, NC, USA

Project: Investigation of protein participation in the interstitial lumen of zebrafish (<https://doi.org/10.1242/dev.199994>)

- Applied Knock-in Tagging methods to examine protein participation in packaging and transporting nutrients in the interstitial lumen of zebrafish through embryo injection, electrophoresis gels, PCR, and fluorescent microscopy screening
- Handling of live zebrafish for tail clips for mutant line maintenance
- Maintained laboratory stocks and upheld standard microbiology aseptic techniques

Rutgers Waksman WISE Student Scholar

July 2019 – August 2019

New Brunswick, NJ, USA

- Authored 4 novel DNA sequences directly isolated and analyzed from *Lemna minor* through PCR, Restriction Enzyme Digests, Gel Electrophoresis, and bioinformatic tools

CLINICAL EXPERIENCE

St. Luke's Hospital <i>Shadowing Volunteer</i> Phillipsburg, NJ, USA	June 2018 - July 2018
• Observed 60 hours of nursing procedures, emergency department patient retrieval systems and gained an understanding of the importance of biomedical technology and software systems	

COMPUTATIONAL EXPERIENCE

DKU Division of Natural and Applied Sciences <i>Student Worker</i> Kunshan, Jiangsu, China	May 2023 – Dec. 2023
• Updated and maintained the DNAS departmental website, implementing user feedback-driven improvements to interface design, page load efficiency, and site structure using WordPress, Python, HTML/CSS, and JS	

FIELD EXPERIENCE

Herpetology <i>Field Course Participant</i> Durham, NC, USA	January 2024 – May 2024
• Conducted field surveys in Duke Forest, monitoring salamander pitfall traps and cover board transects as part of a study on spotted salamanders (<i>Ambystoma maculatum</i>). • Collected and recorded data on species presence, abundance, and habitat conditions as part of a long-term ecological study.	
Certified Advanced Open Water Scuba Diver	August 2017 – Present
• Advanced Open Water certified diver with 30 logged dives (10 shore, 20 boat, 3 night) across reef and coastal environments • Dive Locations: Great Barrier Reef, Costa Rica, Orchid Island (Taiwan), Hawaii, Dutch Springs (Pennsylvania)	
Raritan Headwaters Association <i>Volunteer</i> Raritan, NJ, USA	May 2016 – August 2021
• Conducted quarterly water quality monitoring across 20+ stream sites in the Raritan Headwaters watershed • Performed chemical sampling (dissolved oxygen, nitrates, pH, phosphates) using field kits and standardized protocols • Led ecological surveys to assess stream health through identification and diversity analysis of aquatic macroinvertebrates	

LEADERSHIP EXPERIENCE

Hunterdon County Parks & Recreation <i>Camp Counselor</i> Annandale, NJ, USA	June 2018 – July 2018
• Assisted children with daily activities such as signing in, multiple activities, and breaks	
Murray Hills High School <i>Chinese TA</i> Warren Hills, NJ, USA	August 2016 – May 2017
• Acted as a teacher's assistant for children in a 4th grade Chinese weekend Chinese School	
Golden Lions Marching Band <i>Drum Major and Section Leader</i> Annandale, NJ, USA	August 2017 – June 2021
• Lead and conducted the high school marching band for camp and competitions • Taught new students how to march, play their instruments, and w...	

AI FREELANCE EXPERIENCE

Outlier AI <i>Freelance AI Employee</i> Remote	May 2024 – Sep. 2025
• Worked with Outlier AI on 5 distinct projects for evaluating and creating training data on LLM responses • Specialized in responses with required knowledge in Biology, Python, or Chinese language requirements	

CUSTOMER SERVICE EXPERIENCE

Clinton Station Diner <i>Server and Manager</i> Clinton, NJ, USA	Aug. 2020 – August 2022
• Provided customers with excellent waitstaff	

LABORATORY SKILLS

Wet-lab techniques: Cell culture, Western Blot, PCR/qRT-PCR, Agarose gel electrophoresis, Bacterial transformation, DNA/RNA extraction and purification, siRNA extraction and purification, plasmid design, plasmid and siRNA transfection, sgRNA design, CRISPR/Cas9 knockdown, confocal microscopy, fluorescence microscopy	Dry-lab software program proficiencies and techniques: ImageJ/Fiji, Linux in Terminal and VSCode, SnapGene Viewer, Addgene, CHOPCHOP, PuTTY, BWT, FM-index, FASTA and BLAST heuristics, Phylogenetic Trees with UPGMA and NJ, Cellpose, StarDist, Ilastik, CellProfiler, DeepCell, SAM, AlphaFold 3
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ADDITIONAL SKILLS AND CERTIFICATIONS

- **Programming/Tools:** R/RStudio, Python, Java, HTML/CSS, JS, Linux, MATLAB, Microsoft Office Suite, Colab, Blender, Flask, Docker
- **UX/Design:** Web design, web testing, feedback integration, front-end and back-end debugging
- **Certifications:** CPR & First-Aid Certified (Red Cross), Animal Personnel Annual Training, Hazard Awareness for Animal Facilities, Animal Handling Qualifications (IACUC), Advanced Open Water Diver (PADI)
- **Languages:** Native English Speaker, Fluent in Mandarin and Advanced literacy in Traditional and Simplified Chinese